Dell Unisphere for PowerMax 10.0.0

Installation Guide

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DELL Technologies

Notes, cautions, and warnings

NOTE: A NOTE indicates important information that helps you make better use of your product.

CAUTION: A CAUTION indicates either potential damage to hardware or loss of data and tells you how to avoid the problem.

WARNING: A WARNING indicates a potential for property damage, personal injury, or death.

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About this content

As part of an effort to improve its product lines, Dell periodically releases revisions of its software and hardware. Therefore, some functions that are described in this document might not be supported by all versions of the software or hardware currently in use. The product release notes provide the most up-to-date information on product features.

Contact your Dell technical support professional if a product does not function properly or does not function as described in this document.

This document was accurate at publication time. New versions of this document might be released on the Dell Online Support website. Check the Dell Online Support site https://dell.com/support to ensure that you are using the latest version of this document.

Purpose

This document describes how to configure and use Unisphere for PowerMax.

Related documentation

The following publications provide additional information:

- *Dell Unisphere for PowerMax Release Notes*
- *Dell Unisphere for PowerMax Online Help*
- *Dell Solutions Enabler Installation Guide*
- *Dell Solutions Enabler Release Notes*
- *Dell Solutions Enabler SRM CLI Product Guide*
- *Dell Solutions Enabler CLI Command Reference*
- *Dell PowerMax Family Security Configuration Guide*

Dell technical support, documentation, release notes, software updates, or information about Dell products can be obtained at https://dell.com/support (registration required).

Where to get help

Dell support, product, and licensing information can be obtained as follows:

Product information	Dell technical support, documentation, release notes, software updates, or information about Dell products can be obtained at https://dell.com/support (registration required) or https://www.dell.com/en-us/dt/documentation/vmax-all-flash-family.htm.
Technical support	To open a service request through the Dell Online Support (https://dell.com/support) site, you must have a valid support agreement. Contact your Dell sales representative for details about obtaining a valid support agreement or to answer any questions about your account.

Your comments

Your suggestions help improve the accuracy, organization, and overall quality of the documentation. Send your comments and feedback to: powermaxcontentfeedback@dell.com

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Pre-installation considerations

Topics:

- Before you begin
- Environment and system requirements

Before you begin

The following sections contain steps to take and information to consider before beginning the installation:

Unisphere Initial Setup User

While installing the software, you are prompted to specify the Unisphere Initial Setup User. You can specify the user ID that you use to access the system (if User Authorization is enabled), or you can use the default user ID. If you plan on using the default ID, `smc`, there must be no user IDs listed in the user authorization database. If there are user IDs in the database (regardless of whether user authorization is enabled), you must use SYMCLI to add the default user ID (`smc`) to the database.

For more information about users and roles, see the *Dell PowerMax Family Security Configuration Guide*.

The role of the Unisphere Initial Setup User is to install and set up the Unisphere environment (create users and add roles). It is intended to be a temporary role as it enables you only to perform administrative tasks on storage systems that do not have defined user roles (authorization rules). After an Administrator or a SecurityAdmin is assigned to a storage system, the Unisphere Initial Setup User can no longer access or view the system from the Unisphere console. Dell Technologies recommends that users do not operate in this role for any longer than necessary.

Local, remote, and embedded installation options

Unisphere for PowerMax can be installed in local, remote, or embedded configurations.

In a local configuration, install the Unisphere for PowerMax software on a server that is running Solutions Enabler, which is attached to storage systems: See the Unisphere server in the following image:

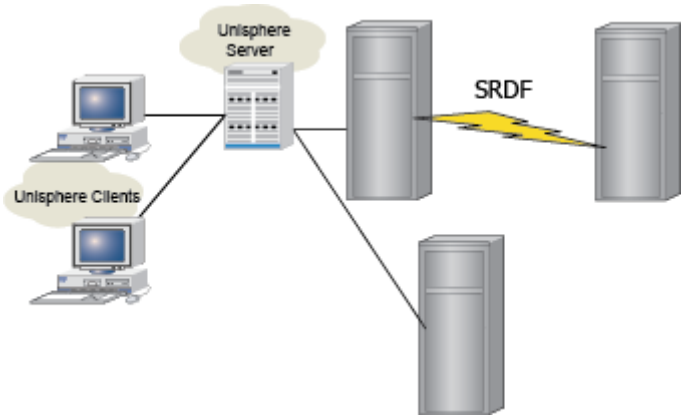


Figure 1. Local installation: Unisphere for PowerMax

In a remote configuration, install the Unisphere software on a server that is connected to the SYMAPI server. See the Unisphere server in the following image:

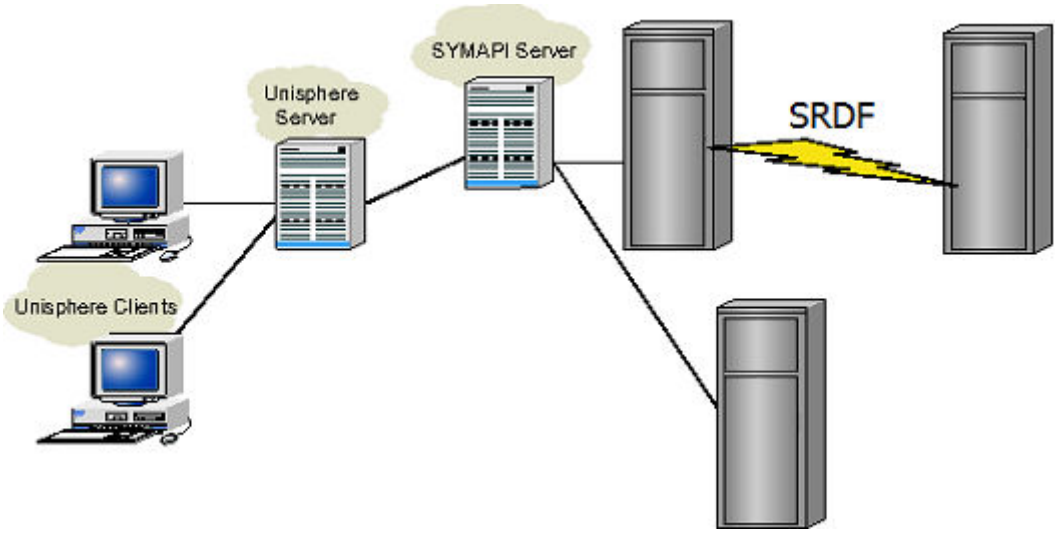


Figure 2. Remote installation: Unisphere for PowerMax on a server connected to a remote SYMAPI server

Alternatively, you can manage storage systems running HYPERMAX OS 5977 or later using the embedded management container application. Embedded management embeds management software (Solutions Enabler, SMI-S, Unisphere for PowerMax) on the array, enabling management of the array without requiring software installed on a host.

Authentication

Unisphere for PowerMax supports the following types of authentication:

Windows (local and domain-based)	Users log in by specifying the Windows domain, username, and password (if they have a Windows account on the SMAS server).
LDAP	Users log in with their LDAP-SSL username and password (if they have a user account that is stored on an LDAP-SSL server). To use this method, a Unisphere Administrator or SecurityAdmin sets up LDAP-SSL authentication in Unisphere for PowerMax. The *Unisphere for PowerMax Online Help* contains instructions on performing these tasks.
Local Unisphere users	Users log in with their Unisphere username and password (if they have a local Unisphere user account). To use this method, a Unisphere Initial Setup User, Administrator, or SecurityAdmin creates a local Unisphere user account for the user. Local user accounts are stored locally on the SMAS server host and work in a similar way as the other methods used to validate user credentials. The *Unisphere for PowerMax Online Help* contains instructions on performing these tasks.
X.509 certificate-based user authentication	Certificate-based user authentication using X.509 certificates is supported on fresh installations on Windows and Linux servers. You can use a certificate, issued by a trusted public third-party certificate authority (CA), to authenticate your identity when using the Unisphere for PowerMax web client. Also, you can use digital identity smartcards, such as Common Access Card (CAC) and Personal Identity Verification (PIV), as part of a multifactor authentication process. You can enable certificate-based user authentication as part of the installation and after you have confirmed your intentions, the choice becomes irreversible. If you select X.509 at installation time, that authentication method and the source of user identity that you configure are irrevocable. You must import the CA certificates before certificate-based user authentication can be used.

The *Dell EMC PowerMax Family Security Configuration Guide* provides more details about authentication, authorization, and other issues related to security.

Environment and system requirements

The Unisphere web server is supported on the platforms that are listed in Server requirements and limitations. Unisphere for PowerMax can be used in remote client/server mode to communicate with SYMAPI servers on other systems, including supported versions of AIX, Linux, Solaris, HP-UX, z/OS, and Windows.

Network requirements

Before installing Unisphere for PowerMax, verify that the host is in a properly configured IP network and that it can resolve IP addresses and hostnames.

When you are installing Unisphere for PowerMax, the use of IPv6 addresses is supported.

Solutions Enabler

Solutions Enabler 64-bit 10.0.0 is required in local and client/server modes. In client/server mode, the `storsrvd` daemon must be running on the server.

Supported PowerMaxOS and HYPERMAX OS versions

Unisphere for PowerMax 10.0.0 supports:

- Storage systems running PowerMaxOS 10 (6079)
- Storage systems running PowerMaxOS 5978
- Storage systems running HYPERMAX OS 5977

Host systems and storage operating environment

Unisphere for PowerMax runs on a wide range of operating systems and works with certain HYPERMAX OS, and PowerMaxOS versions. For detailed interoperability information, search for E-Lab Navigator at https://elabnavigator.dell.com/.

Concurrent users

Unisphere for PowerMax supports up to ten concurrent users, with some performance degradation occurring with more than five storage admin users.

Note the following recommendations on REST API client limits, which must be adhered to in order to achieve optimal performance:

- A maximum of 5 REST API clients can connect to the Unisphere server concurrently.
- A maximum of 50 GET calls can be made to the Unisphere server per second. These calls can be made up of performance REST API GET and POST calls and nonperformance REST API GET calls.
- A maximum of 5 nonperformance POST or PUT calls can be run against the Unisphere server concurrently.
- It is recommended that you use bulk REST API calls, where possible.

NOTE: If the recommendation above are not followed, it might result in performance degradation of the Unisphere for PowerMax server.

VMware and Hyper-V guests

Unisphere for PowerMax is supported on a guest operating system on the following platforms:

- VMware ESX/ESXi Server versions 6.5, 6.7, and 7.0 (and all updates of each version)
- Windows Server 2019
- Virtual machines must not run on shared memory or resources.

Ensure that the following conditions are met:

- The guest operating system is listed in the Dell E-lab Navigator and in VMware/Hyper-V support documentation.
- Unisphere for PowerMax supports the platform.

NOTE: The guest operating systems must provide the same CPU, memory, disk, gatekeeper, and other requirements as if Unisphere for PowerMax were installed on a physical machine.

Client screen requirements

The minimum client screen resolution that is required is 1280 x 1024.

Client browser requirements

The Unisphere for PowerMax client is viewed in a browser and does not use dynamic ports, so it works with most VPN solutions.

NOTE: The preferred browsers for Unisphere for PowerMax are Chrome and Firefox.

The following table defines the client browser requirements:

Table 1. Unisphere for PowerMax client browser requirements

Browser	Version	Memory [a]	Platform
Chrome	102	600 MB	Windows
Firefox	91		
Microsoft Edge	102		

a. Required for the browser as it runs on the machine from which the console is launched.

Server requirements

The table defines the server requirements for Unisphere for PowerMax.

Table 2. Server requirements and limitations

Operating system[a]	Maximum volume count	Minimum processor specification	Minimum available memory[b]	Minimum available disk space	Maximum managed storage system count
Windows	0–64,000	Minimum of a two-core, 2.5 GHz processor	12 GB	120 GB	10
Linux					
Windows	64,001–128,000	Minimum of a two-core, 2.5 GHz processor	16 GB	140 GB	15
Linux					
Windows	128,001–256,000	Minimum of a six-core, 2.5 GHz processor	24 GB	180 GB	20
Linux					

a. For specific operating system information, see E-Lab™ Interoperability Navigator at: https://elabnavigator.dell.com/
b. The minimum amount of memory that is required on the server (before installation) to successfully run Unisphere for PowerMax. It is not the total amount of memory on the server. If a server has 12 GB of physical memory that is installed but there is only 6 GB available before Unisphere for PowerMax is installed, you may experience performance issues with the application. If running on a Virtual Machine, this amount must be dedicated memory, not shared memory.

Other requirements

- DNS must be configured correctly on the host.
- The host system must have connectivity to one or more storage systems to add users and set permissions in Unisphere.
- Unisphere for PowerMax can be installed on the same system as the SYMAPI server, or on a system connected to the SYMAPI server. For more information, see Local installation: Unisphere for PowerMax and Remote installation: Unisphere for PowerMax on a server connected to a remote SYMAPI server, respectively.
- To install PostgreSQL on Linux, a minimum of 2 GB of shared memory must be available.

For more information about how to configure the size of a shared memory segment, using the `shmmax` kernel parameter, see the documentation for your operating system.

- If you are changing your SYMAPI connection type, such as changing from a local connection to a client/server configuration, you must perform a fresh installation.
- Before installing Unisphere for PowerMax on a host running Linux, ensure that the maximum number of open file descriptors that are allowed is at least 4096.
- For installations on Microsoft Windows, Unisphere must be installed on a volume which has Windows short file names (8dot3) enabled.

  To determine if short names are enabled for a volume, use:

  ```
  fsutil 8dot3name query Volume_Drive_Letter
  ```

  Return values are:

  Volume state = 0 (8dot3 is enabled)

  Volume state = 1 (disabled)

  To enable 8dot3, use:

  ```
  fsutil 8dot3name set Volume_Drive_Letter 0
  ```

  For example, to enable 8dot3 on the C drive

  ```
  fsutil 8dot3name set C: 0
  ```

- Ensure that your system is configured to use a supported operating system, language, and keyboard.
- Before installing Unisphere on a Microsoft Windows system, ensure that the language for the system location is correct to avoid indecipherable characters on the install screens. To check the language, go to **Control Panel** > **Region and Language** > **Administrative tab** > **Language for non-Unicode programs**.
- Unisphere for PowerMax supports Active Directory over LDAP and Open LDAP version 2.4 and later.

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Installing Unisphere for PowerMax

Topics:

- Preinstallation configuration
- Installing Unisphere for PowerMax
- Completing the installation
- Launching Unisphere for PowerMax

Preinstallation configuration

Before you install Unisphere for PowerMax, complete the following tasks if they are appropriate for your installation environment:

- Setting the event and statistics daemons to autostart on Linux
- Setting up the PostgreSQL user on Linux
- Configuring networking on Linux

Setting the event and statistics daemons to autostart on Linux

To install Unisphere for PowerMax on a local Linux 64-bit host, complete this task.

To set the event and statistics daemons to autostart:

1. Set the event and statistics daemons to autostart using the following commands:

```
stordaemon install storevntd -autostart
stordaemon install storstpd -autostart
```

2. Start the same daemons using the following commands:

```
stordaemon start storstpd
stordaemon start storevntd
```

Setting up the PostgreSQL user on Linux

Before starting a new installation of Unisphere for PowerMax on Linux, the PostgreSQL user (`postgres`) and group (`postgres`), must be present. Also, the user must be a member of the `postgres` group.

Use the following commands to create the `postgres` group and add the `postgres` user to the `postgres` group:

```
groupadd postgres
useradd -g postgres postgres
```

Configuring networking on Linux

In the `/etc/hosts` file, comment out the following text:

```
::1
```

This change is needed for both IPv4 and IPv6 configurations.

Configuring file system mountpoints on Linux systems

About this task

When mounting the `/tmp` directory, you must remove the `noexec` parameter.

If you do not want to remove the `noexec` parameter for the `/tmp` directory, set an alternative directory for the installer.

For `sh`, `ksh`, `bash`, and `zsh`:

```
$ IATEMPDIR=new_temp_dir
$ export IATEMPDIR
```

For `csh` and `tcsh`:

```
$ setenv IATEMPDIR new_temp_dir
```

Installing Unisphere for PowerMax

Prerequisites

NOTE: Before installing Unisphere for PowerMax 10.0.0, you must install Solutions Enabler 10.0.0 and run the `symcfg discover` command.

About this task

To install Unisphere for PowerMax:

Steps

1. Accessing the software: Step 1: Accessing the software .
2. Installing the software: Step 2: Installing the software.
3. Installing Unisphere for PowerMax using the wizard: Step 2, option 1: Installing Unisphere for PowerMax using the wizard.
4. Installing Unisphere for PowerMax from the command line: Step 2, option 2: Installing Unisphere for PowerMax from the command line.
5. Installing Unisphere for PowerMax using silent install: Step 2, option 3: Installing Unisphere for PowerMax using silent install

Step 1: Accessing the software

About this task

The following Unisphere for PowerMax software and documentation is available on the Dell support website:

- Windows 64-bit software:

  ```
  UNIVMAX_aa.bb.cc.dd_RELEASE_WINDOWS_X86_64.exe
  ```

- For Linux 64-bit:

  ```
  UNIVMAX_aa.bb.cc.dd_RELEASE_LINUX_X86_64.bin
  ```

  In the installation file name:
  - *aa* is the major release number.
  - *bb* is the minor release number.
  - *cc* is the service release number.
  - *dd* is the patch number.

- *Unisphere for PowerMax Online Help*

- *Unisphere for PowerMax Release Notes*
- This installation guide

To access the software:

Steps

1. On the Dell support website:
   a. Search for **`Unisphere for PowerMax`**.
   b. Locate the appropriate installation file and download it to a directory on the host drive.
2. Go to Step 2: Installing the software.

Step 2: Installing the software

About this task

You can install Unisphere for PowerMax in any of the following ways:

- Using an installation wizard, as described in Step 2, option 1: Installing Unisphere for PowerMax using the wizard
- From the command line, as described in Step 2, option 2: Installing Unisphere for PowerMax from the command line
- Using silent install, as described in Step 2, option 3: Installing Unisphere for PowerMax using silent install

Step 2, option 1: Installing Unisphere for PowerMax using the wizard

Prerequisites

To enable X.509 certificate-based authentication, you need an appropriate X.509 smart card reader. You must have access to any CA certificates required for authentication. All CA certifications must be imported at installation time. Any certificates that are installed must be less than 32 KB in size.

The *Dell EMC PowerMax Family Security Configuration Guide* provides more details about authentication, authorization, and other issues related to security.

About this task

To install Unisphere for PowerMax using the wizard:

Steps

1. Save all files and exit all applications.
2. Change the directory to the location of the kit and run the appropriate executable file for your operating system.
   To install Unisphere for PowerMax using the wizard in Linux, enable the display manager and use the `-i swing` switch, for example:

   ```
   ./UNIVMAX_aa.bb.cc.dd_RELEASE_LINUX_X86_64.bin -i swing
   ```

   If the display manager is not enabled or the `-i swing` switch is not used, the installer defaults to console mode.

   If you are using the nonroot user feature on a Linux installation, the Unisphere installer picks up the username running the SE daemons and assigns that to the Unisphere Java process.
3. On the **Introduction** page, review the recommendations and click **Next**.
4. On the **Choose Install Folder** page, click **Next** to accept the default directory, or click **Choose** to select another directory.

   NOTE: On Red Hat Enterprise Linux 8.0 and above, installing on a nonroot partition is not supported.

5. On the **Symapi Configuration** page:
   a. Specify a **Unisphere Initial Setup User** name (default is `smc`). If you have enabled authorization on the Unisphere server host, you can either accept the default or use your Windows username. For more information about the role of the Unisphere Initial Setup User, see Unisphere Initial Setup User.
   b. Specify a **SymAPI Connection Type** for the initial login based on the following:

- **Local**—On a host that has Solutions Enabler installed and is attached to PowerMax systems.
- **Remote**—On a remote server that is connected to a SYMAPI server, specify the SYMAPI server **Node Name** and **Net Port**. The default net port for SYMAPI is 2707.

For more information about each of the connection types, see Local, remote, and embedded installation options.

c. Click **Next**.

6. On the **Ports Configuration** page, specify the ports to use or enable when connecting to the Unisphere server.
The default ports of the Unisphere server (**HTTPS Port No**) and performance database (**DB Port No**) are shown. To use the default ports, verify that they are available, and click **Next** to accept and enable them. Or you can specify and enable different ports, and click **Next**.
7. Optional: On the **X.509 Certificate-based Client Authentication** page, configure certificate-based user authentication options. The default behavior is for X.509 certificate-based authentication to remain disabled.

   To enable X.509 certificate-based authentication, complete the following steps:

   a. Select **Enable certificate based client authentication?**.
   b. To enable X.509 certificate-based authentication, you must import at least one CA certificate. To do this, complete the following steps:
      i. Select **Import Trusted CA cert (Certificate)**.
      ii. Click **Next**.
      iii. Click **Choose File**.
      iv. On Windows systems, browse to and select one or more certificate files. On Linux systems, type the path to the certificate.
      v. In the confirmation dialog box, click **Yes**.
   c. Click **Next**.
   d. On the **Confirm Client Authentication** dialog box, click **OK**.
8. To install Unisphere for PowerMax on a host with more than one network interface configured, you are prompted to choose the IP address to which Unisphere for PowerMax binds. On the **Listening (bind) Address Selection** page, select the server protocol (IPv4 or IPv6), select the IP address, and then click **Next**.
9. If the requirements for disk space, or memory, or both, are not met, a warning is displayed. If you want to continue the installation, regardless of the consequences that are described in the message, click **OK**.
10. On the **Pre-Installation Summary** page, verify the summary information, and click **Install** to continue the installation.
11. On the **Install Complete** page, click **Done**.
12. Go to Completing the installation for instructions on completing the installation.

Step 2, option 2: Installing Unisphere for PowerMax from the command line

Prerequisites

To enable X.509 certificate-based authentication, you need an appropriate X.509 smart card reader. You must have access to any CA certificates required for authentication. All CA certifications must be imported at installation time. Any certificates that are installed must be less than 32 KB in size.

The *Dell EMC PowerMax Family Security Configuration Guide* provides more details about authentication, authorization, and other issues related to security.

About this task

When you are installing Unisphere for PowerMax from the command line, online help is available by entering `-?`.

To install Unisphere for PowerMax from the command line:

Steps

1. Save all files and exit all applications.
2. Change the directory to the location of the installation file.
3. Run the following command to install Unisphere for PowerMax on Windows:

```
installation_file -i console
```

Run the following commands to install Unisphere for PowerMax on Linux:

```
chmod +x installation_file
./installation_file -i console
```

Where *installation_file* is one of the following operating system-specific installation file names:

- For Windows 64-bit:

  ```
  start /wait UNIVMAX_aa.bb.cc.dd_RELEASE_WINDOWS_X86_64.exe
  ```

  The start /wait option starts the installer and waits for it to end.
- For Linux 64-bit:

  ```
  UNIVMAX_aa.bb.cc.dd_RELEASE_LINUX_X86_64.bin
  ```

In the installation file name:

- `aa` is the major release number.
- `bb` is the minor release number.
- `cc` is the service release number.
- `dd` is the patch number.

If you are using the nonroot user feature on a Linux installation, the Unisphere installer picks up the username running the SE daemons and assigns that to the Unisphere Java process.

4. On the **Introduction** panel, review the recommendations and press **Enter**.
5. On the **Choose Install Folder** panel, press **Enter** to accept the default directory, or specify a different directory and press **Enter**.

   NOTE: On Red Hat Enterprise Linux 8.0 and above, installing on a nonroot partition is not supported.

6. On the **postgres User and Group** panel, you are instructed to create the `postgres` group and `postgres` user before continuing. When complete, press **Enter**.
7. On the **Initial Setup User and SymAPI Configuration** panel, complete the following steps:
   a. Specify a **Unisphere Initial Setup User name** (default is `smc`).

      You can either accept the default username or if you have enabled authorization on the Unisphere server host, use your Windows username. For more information about the role of the Unisphere Initial Setup User, see Unisphere Initial Setup User.
   b. Specify a **SymAPI Connection Type** for the initial login based on the following:
      - **Local**—On a host with Solutions Enabler installed and attached to PowerMax systems
      - **Remote**—On a remote server that is connected to a SYMAPI server, specifies the SYMAPI server **Node Name** and **Net Port**. The default net port for SYMAPI is 2707.

      For more information about each of the connection types, see Local, remote, and embedded installation options.
8. On the **Ports Configuration** panel, specify the ports to use or enable when connecting to the Unisphere server.
   The default ports of the Unisphere server (**HTTPS Port No**) and performance database (**DB Port No**) are shown. To use the default ports, verify that they are available, and click **Next** to accept and enable them. Or you can specify and enable different ports, and click **Next**.
9. Optional: On the **X.509 Certificate-based Client Authentication** panel, configure certificate-based user authentication options. The default behavior is for X.509 certificate-based authentication to remain disabled.

   To enable X.509 certificate-based authentication, complete the following steps:
   a. At the **Enable certificate based client authentication?** prompt, type **2** for Yes.
   b. To enable X.509 certificate-based authentication, you must import at least one CA certificate. To do this, complete the following steps:
      i. At the **Do you want to Import trusted CA cert(Certificate) ?** prompt, type **2** for Yes.
      ii. When prompted to do so, enter the absolute path to the certificate file.
      iii. If you want to install another certificate, at the **Do you want to Import trusted CA cert(Certificate) ?** prompt, type **2** for Yes, otherwise type **1** to continue the installation.
      iv. A summary of the certificates to be installed is displayed.
   c. Press **Enter** to continue.

d. On the **Confirm Client Authentication** panel, confirm your selection.

10. On the **Listening(bind) Address Selection** panel, to install Unisphere for PowerMax on a host with more than one network interface configured, choose the IP address to which Unisphere for PowerMax binds. You can specify if it is an IPv4 or an IPv6 address.
11. If the requirements for disk space, or memory, or both are not met, a warning is displayed. If you want to continue the installation, regardless of the consequences that are described in the message, type **1**.
12. On the **Pre-Installation Summary** panel, verify the summary information, and the press **Enter** to continue the installation.
13. On the **Install Complete** panel, press **Enter**.
14. Go to Completing the installation for instructions on completing the installation.

Step 2, option 3: Installing Unisphere for PowerMax using silent install

About this task

Silent install provides a noninteractive mode of installation using a response file. It enables installation simultaneously on multiple hosts and helps to automate the install process.

Typical response file contents are in the form of key-value pairs. The installer interprets the keys in the response files as InstallAnywhere variables and replaces their values at runtime. Typical contents of a response file are as follows:

```
EMC_SUNAME=smc
EMC_HTTPSPORT= [PORT NUM]
EMC_DBPORT=[PORT NUM]
EMC_CONNTYPE=[Local | Remote]
EMC_NODENAME=[localhost| remote host]
EMC_NETPORT=[PORT NUM]
BIND_ADDRESS=[IP ADRESS]
IS_RESPONSE_FILE=[always true]
USER_INSTALL_DIR=[install directory]
```

The following list contains definitions of the variables:

EMC_SUNAME	Initial Setup Username
EMC_HTTPSPORT	Custom HTTPS port value
EMC_DBPORT	Database port value
EMC_CONNTYPE	SYMAPI connection type. Can be either Local or Remote.
EMC_NODENAME	Sets the IP of the host on which SYMAPI is running.
EMC_NETPORT	Port value for Remote SYMAPI connection type.
BIND_ADDRESS	The bind address used by JBoss
IS_RESPONSE_FILE	Mandatory variable to inform the installer that the response file is passed in silent install mode and input should be taken from the response file.
USER_INSTALL_DIR	Installation path

To install Unisphere for PowerMax using silent install:

Steps

1. In the install directory, create a response file with a `.properties` extension and add the parameters that are listed above. The following example shows a response file with default options:

```
EMC_SUNAME=smc
EMC_HTTPSPORT=8443
EMC_DBPORT=3324
EMC_CONNTYPE=Local
```

```
EMC_NODENAME=localhost
EMC_NETPORT=2707
BIND_ADDRESS=198.51.100.99
IS_RESPONSE_FILE=TRUE
USER_INSTALL_DIR=C:\\Program Files\\EMC
```

NOTE: On Linux, the install directory is `/opt/emc`.

NOTE: On Red Hat Enterprise Linux 8.0 and above, installing on a nonroot partition is not supported.

2. Invoke a silent install from the command line by typing one of the following commands:
   - For Linux:

     ```
     ./UNIVMAX_aa.bb.cc.dd_RELEASE_LINUX_X86_64.bin -i silent [-f response_file]
     ```

   - For Windows:

     ```
     UNIVMAX_aa.bb.cc.dd_RELEASE_WINDOWS_X86_64.exe -i silent [-f response_file]
     ```

   In the installation file name:
   - *aa* is the major release number.
   - *bb* is the minor release number.
   - *cc* is the service release number.
   - *dd* is the patch number.

   NOTE: For Windows you can use `Start /wait` to end the command execution:

   ```
   start /wait UNIVMAX_aa.bb.cc.dd_RELEASE_WINDOWS_X86_64.exe -i silent -f response_file
   ```

Completing the installation

About this task

This section describes the tasks to perform after the installation has been completed:
- Configuring language and regional settings: Configuring language and regional settings
- Configuring third-party utilities: Configuring third-party utilities
- Starting and stopping the performance database: Starting and stopping the performance database
- Starting and stopping the SMAS service: Starting and stopping the SMAS service

Configuring language and regional settings

About this task

For Windows users outside the United States, you must configure your regional and language settings to English (United States) if you plan on using the graph feature of the Quality of Service (QoS) and Replication Monitors.

The exact steps that you must follow might vary from the steps that are listed here, depending on the version of Windows you are using.

To configure the language and regional settings:

Steps

1. Stop the SMAS service (if it is running). See Starting and stopping the SMAS service.
2. In the Windows **Control Panel**, double-click **Regional and Language Options**.
3. On the **Regional Options** tab, set the **Standards and formats to English (United States)**.
4. On the **Advanced** tab, set the **Language for non-Unicode programs to English (United States)** and select **Apply all setting to the current user account and to the default user profile**.
5. Restart the service.

Configuring third-party utilities

About this task

To ensure that third-party software utilities (including virus scanning and backup tools) do not affect the operation of the Unisphere for PowerMax software that is installed on the host, you might need to carry out some additional configuration.

Exclude the `install_dir\SYMAPI\stp\spa` directory and all subdirectories from any virus scan.

Exclude the performance database directory, `install_dir\SMAS\jboss\standalone\data\pgsql\data`, and all subdirectories, from the actions performed by any third-party software utility.

NOTE: Not following this advice may lead to data corruption in the performance database or problems with data collection.

Starting and stopping the performance database

About this task

If required to do so, you can start and stop the performance database manually.

Windows

About this task

To start or stop the EMC_smasdb service in Windows, use the Control Panel (**Administrative Tools** > **Services**).

Alternatively, you can use the following commands to start or stop the Windows services using the CLI:

```
net start "EMC_smasdb"
net stop "EMC_smasdb"
```

Linux

About this task

Start the performance database before starting the SMAS service. In the same way, stop the SMAS service before stopping the performance database.

To start or stop the performance database (PostgreSQL) in Linux, go to this directory: `install_dir/SMAS/jboss/standalone/data/pgsql` and use the following commands:

Action	Command
Start	`./smasdb_unix_helper.sh -S`
Stop	`./smasdb_unix_helper.sh -k`

Starting and stopping the SMAS service

About this task

If required to do so, you can start and stop the SMAS service manually.

Windows

About this task

To start or stop the SMAS server in Windows, use the **Services** panel. Click **Control Panel** > **Administrative Tools** > **Services** to open the **Services** panel.

Alternatively, you can use the following commands to start or stop the Windows services using the CLI:

```
net start "EMC Symmetrix Management Application Server"
net stop "EMC Symmetrix Management Application Server"
```

Linux

About this task

Start the performance database before starting the SMAS service. In the same way, stop the SMAS service before stopping the performance database.

To start or stop the SMAS service in Linux, change to this directory: `/etc/init.d` and use the following commands:

Action	Command
Start server	`./smas start`
Stop server	`./smas stop`

Launching Unisphere for PowerMax

About this task

NOTE: After performing any of the following operations, it can take up to 30 minutes (depending on the size of the storage environment) before Unisphere for PowerMax becomes available:

- Install Unisphere for PowerMax
- Upgrade Unisphere for PowerMax
- Reboot
- Restart of the SMAS service

To launch Unisphere for PowerMax:

Steps

1. Type one of the following URLs in a browser:

   ```
   https://host_IP:port_number
   ```

   or

   ```
   https://host_name:port_number
   ```

   The port number is configured during installation. The default HTTPS port number is 8443.

   If the host IP address is an IPv6 address, surround the IP address with square brackets, for example:

   ```
   https://[2001:db8:ffff:ffff:ffff:ffff:ffff:ffff]:8443
   ```

   If the host IP address is an IPv4 address, type the IP address as normal, for example:

   ```
   https://198.51.100.255:8443
   ```

2. Do one of the following:
   - At the login window, type the Unisphere Initial Setup User username and password, and click **Login**.

     The default username for the Unisphere Initial Setup User is `smc` and the default password is `smc`.

   - If you are using certificate-based user authentication, click **Options** to display more options, and select **Use Client Certificate**. If the Unisphere for PowerMax login window does not display correctly, ensure that the CA certificates have been imported correctly.

3. Click **Login**.

3

Upgrading Unisphere for PowerMax

Topics:

- Upgrade paths
- Before upgrading Unisphere for PowerMax
- Running the Unisphere for PowerMax upgrade
- After upgrading Unisphere for PowerMax

Upgrade paths

You can upgrade directly to Unisphere for PowerMax 10.0.0 from Unisphere for PowerMax 9.2.0, or later.

NOTE: To upgrade to 10.0.0 from a version earlier than 9.2.x, first upgrade to 9.2.x, and then to 10.0.0.

NOTE: Unisphere for PowerMax release 10.0.0 onwards uses PostgreSQL version 12.

On SUSE Linux Enterprise Server platforms, PostgreSQL version 12 is supported only on SUSE Linux Enterprise Server version 12.5 or later.

On Red Hat Enterprise Linux Server platforms, PostgreSQL version 10 is supported only on Red Hat Enterprise Linux Server version 7.8, or later.

The installation program automatically upgrades your installation according to the settings used in the previous installation, including port and connection type.

The following Unisphere for PowerMax settings from your previous installation are maintained through the upgrade process:

- Users
- Alert policies and threshold settings
- Email settings: Mailing list and SMTP configuration
- Performance dashboard settings
- Queries

Before upgrading Unisphere for PowerMax

Prerequisites

NOTE: Before upgrading to Unisphere for PowerMax 10.0.0, you need Solutions Enabler 10.0.0.

About this task

Ensure that the minimum RAM and disk space requirements for Unisphere for PowerMax are met. For more information about server requirements, see Pre-installation considerations.

If upgrading Solutions Enabler on a Linux platform from a version lower than 9.2.x, you must perform the upgrade only as the root user (that is, the user running the Solutions Enabler daemons). This behavior is because the root user installed Solutions Enabler 9.1. Attempting to perform the Solutions Enabler upgrade as a nonroot user fails.

Upgrades to Unisphere for PowerMax are supported from root to root, or nonroot to nonroot only. The Unisphere for PowerMax Java process runs using the same user account that runs the Solutions Enabler daemons.

Before commencing any upgrade process, ensure that backups of all data have been made.

If any LocalDir accounts have £ or € characters in the password, you must change the password before proceeding with the upgrade to Unisphere for PowerMax 10.0.0.

As a safeguard, follow these steps:

Steps

1. Back up the system database and for each registered storage system, back up the performance database.
2. Take note of the systems that are registered for collecting performance data. Record the settings for data collection and the diagnostic interval for each system.
3. Export the Performance settings (metric and alert definitions and user templates). This operation requires setting a password to use when you import the saved settings to the new system.

Running the Unisphere for PowerMax upgrade

About this task

This section describes how to upgrade the Unisphere for PowerMax software.

Upgrading Unisphere for PowerMax on Windows

About this task

To upgrade Unisphere for PowerMax using the wizard on Windows:

Steps

1. Access the software, as described in Step 1: Accessing the software.
2. Go to the location of the installation files and run the appropriate executable file for your operating system.
3. In the **Unisphere for PowerMax Instance Detected** dialog box, click **Yes** to continue and upgrade.
4. A warning about support for V2 arrays is displayed. Click **OK** to continue the upgrade.
5. A database backup warning is displayed. Click **OK** to continue the upgrade. To preserve performance data, ensure that a backup has been performed.
6. In the **Introduction - Upgrade Unisphere for PowerMax** page, click **Next**.
7. In the **Pre-Installation Summary** page, click **Install**.
8. When the upgrade process completes, click **Done**.

Upgrading Unisphere for PowerMax on Linux

About this task

To upgrade Unisphere for PowerMax using the command line on Linux:

Steps

1. Stop the SMAS service, as described in Starting and stopping the SMAS service.
2. Access the software, as described in Step 1: Accessing the software.
3. Go to the location of the installation files and run the appropriate executable file for your operating system.
4. When prompted to do so, confirm that you want to upgrade the installed features.
5. The installation script displays the current Unisphere version and the version to which you are upgrading. Press **Enter**.
6. A warning about support for V2 arrays is displayed. Press **Enter** to continue the upgrade.
7. A database backup warning is displayed. Press **Enter** to continue the upgrade. To preserve performance data, ensure that a backup has been performed.
8. In the **Introduction - Upgrade Unisphere for PowerMax** page, press **Enter**.
9. In the **Pre-Installation Summary** page, press **Enter**.
10. Press **Enter** through each option in the upgrade process.
    The upgrade process completes.

Next steps

Complete the tasks described in After upgrading Unisphere for PowerMax.

Upgrading Unisphere for PowerMax using silent install

About this task

Unisphere for PowerMax supports silent install upgrades. During an upgrade, a response file and `-f` options are not required as the installer takes the parameters from previously deployed persistent files. Complete the following steps to perform a silent install upgrade:

Steps

Run one of the following commands:

- For Linux: `UNIVMAX_aa.bb.cc.dd_RELEASE_LINUX_X86_64.bin`
- For Windows: `UNIVMAX_aa.bb.cc.dd_RELEASE_WINDOWS_X86_64.exe`

In the installation file name:

- *aa* is the major release number.
- *bb* is the minor release number.
- *cc* is the service release number.
- *dd* is the patch number.

NOTE: For Windows, you can use `start /wait` to end the command execution: `start /wait UNIVMAX_aa.bb.cc.dd_RELEASE_WINDOWS_X86_64.exe -i silent`

After upgrading Unisphere for PowerMax

About this task

This section describes the tasks to perform after the installation has been completed.

Renewing link and launch client registrations

Steps

If you were using link and launch client registrations to launch Unisphere from ProSphere, or vSphere VSI clients, re-register your host with Unisphere for PowerMax. For more information, see the Unisphere for PowerMax Online Help.

Restoring and verifying system and performance data

About this task

To restore and verify system and performance data, complete the following steps:

Steps

1. Re-register the systems that were collecting performance data before the upgrade.
2. Verify your data collection and diagnostic intervals against the values that are recorded before the upgrade.
3. Restore your performance database backups using the Unisphere for PowerMax console and verify each database.
4. Verify the Performance settings (metric and alert definitions and user templates). If you want to import the Performance settings, use the password that you set when you exported the settings.

   The Unisphere for PowerMax Online Help contains instructions on performing these tasks.

4

Uninstalling Unisphere for PowerMax

Topics:

- Before you begin
- Uninstalling from Windows
- Uninstalling from Linux

Before you begin

Before uninstalling Unisphere for PowerMax, note the following:

- Uninstalling Unisphere for PowerMax removes the performance database. The Unisphere for PowerMax online help provides instructions on backing up and restoring the performance database, should you want to reuse it later.
- During an uninstallation, the installation program prompts you to export your customized performance settings (metrics and threshold settings only) to a file that can be imported to another Unisphere for PowerMax environment. The Unisphere for PowerMax online help provides instructions on exporting and importing customized performance settings.
- Uninstalling Unisphere for PowerMax terminates all client sessions to the SMAS server and any SYMAPI operations in progress.

NOTE: Before you uninstall Unisphere for PowerMax, Dell Technologies recommends that you unregister any storage systems that are registered for performance data.

Uninstalling from Windows

About this task

You can uninstall Unisphere for PowerMax from a Windows host using either the Windows **Add/Remove Programs** window, the command line, or a wizard.

Using Windows Add/Remove Programs

About this task

The exact steps that you must follow, and the names of particular windows, might vary from what is listed here, depending on the version of Windows that you are using.

To uninstall Unisphere for PowerMax from the Windows **Add/Remove Programs** window:

Steps

1. At the Windows command prompt, type `appwiz.cpl` and press **Enter**.
   The window where you can uninstall or change a program is displayed.
2. In the **Add/Remove Programs** dialog, select **Dell EMC Unisphere for PowerMax** and click **Change/Remove**.
3. Complete the remaining steps in the wizard.

Using the Windows command line

About this task

To uninstall Unisphere for PowerMax using the command line:

Steps

1. Change folder location to *install_dir*\SMAS_EMC_SMAS_installation.
2. Type the following to launch the uninstallation console procedure: `Uninstall_EMC_SMAS_Installation.exe`
3. Follow the steps on screen.

   A few files remain after the uninstallation operation. To remove them, delete the `SMAS` folder.

Using the Windows wizard

About this task

To uninstall Unisphere for PowerMax using the wizard:

Steps

1. Change folder location to *install_dir*\SMAS_EMC_SMAS_installation.
2. Launch the wizard: `Uninstall_EMC_SMAS_Installation.exe`
3. Complete the steps in the wizard.

   A few files remain after the uninstallation operation. To remove them, delete the SMAS folder.

Uninstalling using silent uninstall on Windows

About this task

To uninstall Unisphere for PowerMax in silent uninstall mode:

Steps

1. Change the location to `install_dir\SMAS\_EMC_SMAS_installation`.
2. Type the following to launch the uninstallation: `Start /wait Uninstall_EMC_SMAS_Installation.exe -i silent`.

Uninstalling from Linux

About this task

You can uninstall Unisphere for PowerMax from a Linux host using either the wizard or the command line.

Using the wizard

About this task

Using the wizard to uninstall Unisphere for PowerMax is not supported on virtual machines.

To uninstall Unisphere for PowerMax from a Linux host using the wizard:

Steps

1. Go to the following directory: *install_dir*/SMAS/_EMC_SMAS_installation.
2. Type the following to launch the uninstallation wizard: `./Uninstall_EMC_SMAS_Installation -i swing`
3. Complete the steps in the wizard.

   A few files remain after the uninstallation operation. To remove them, delete the SMAS folder.

Using the Linux command line

About this task

To uninstall Unisphere for PowerMax from a Linux host using the command line:

Steps

1. Go to the following directory:

   ```
   install_dir/SMAS/_EMC_SMAS_installation
   ```

2. Type the following to launch the uninstallation console procedure:

   ```
   ./Uninstall_EMC_SMAS_Installation
   ```

3. Follow the steps on screen.

   A few files remain after the uninstallation operation. To remove them, delete the SMAS folder.

Uninstalling using silent uninstall on Linux

About this task

To uninstall Unisphere for PowerMax in silent uninstall mode:

Steps

1. Change the location to `install_dir/SMAS/_EMC_SMAS_installation`.
2. Type the following to launch the uninstallation: `./Uninstall_EMC_SMAS_Installation -i silent`.

5

Appendix A - Technical Notes

Topics:

- Technical notes

Technical notes

This appendix provides technical notes for the advanced configuration of this version of Unisphere for PowerMax.

Managing thin pools

It is recommended that thin pools contain DATA volumes of roughly the same size.

SYMAPI server

If you want to change the SYMAPI server that Unisphere is pointing to, uninstall Unisphere, and then reinstall it.

During the install, you supply the new SYMAPI hostname and network port number.

Local replication management

Unisphere for PowerMax cannot manage replication sessions that are managed exclusively by device files.

Advanced configuration settings

Unisphere provides a dialog box for setting storage system preferences, including debug settings. The debug settings are restricted to Unisphere administrators.

Use the debug settings only at the direction of Dell Customer Service.

Configuring idle user timeout

By default, the client browser logs out automatically after 480 minutes (8 hours). To change this value, complete the following steps:

1. Open the following file for edit:

   ```
   install_dir\jboss\domain\deploy\smc.ear\smc.war\WEB-INF\web.xml
   ```

2. Change the value of the `sageIdleUserTimeoutInMinutes` context parameter to any positive integer, and save the file.
3. If it exists, delete the `install_dir\jboss\domain\deploy0` folder.
4. Restart Unisphere for PowerMax.

Alerts

By default, most Unisphere status alerts are disabled. *Configuring alert policies* in the Unisphere for PowerMax help includes instructions on enabling them.

NOTE: For additional information about event and alerts, see the *Events and Alerts for PowerMax and VMAX Users Guide*.

Backups

Consider making regular backups of the system and performance databases, scheduling backup tasks where appropriate and possible. For more information about managing backups, see the *Unisphere for PowerMax online help*.

Protecting against POODLE SSL v3 vulnerability

To protect your system from the Padding Oracle On Downgraded Legacy Encryption vulnerability, ensure that following is set in your browser:

- SSL v3 protocol is disabled.
- TLS 1.2 protocol is enabled.

For instructions, see the documentation for your browser.

Troubleshooting the "database is in maintenance mode" message

If a database restore, backup, or startup operation is in progress and an attempt is made to start a backup, restore, or registration operation, the following message might be displayed:

```
The following Symmetrix database is in maintenance mode: storage_system_number
```

To resolve this issue, in Unisphere for PowerMax complete the following steps:

1. Click **Performance** > **Settings** > **Databases**.
2. Wait until the **Status** of the database of the appropriate storage system is set to **OK** before attempting to start another database operation.